

Activity-02

29/7/2026

1) Stack and Queue problem.

Stack operations:

- $\text{Push}(54) \rightarrow [54]$
- $\text{Push}(52) \rightarrow [54, 52]$
- $\text{POP}() \rightarrow \text{removes } 52$
- $\text{push}(55) \rightarrow [54, 55]$
- $\text{Push}(62) \rightarrow [54, 55, 62]$
- $S = \text{pop}() \rightarrow S = 62$

Queue operations:-

- $\text{enqueue}(21) \rightarrow [21]$
- $\text{enqueue}(24) \rightarrow [21, 24]$
- $\text{dequeue}() \rightarrow \text{removes } 21$
- $\text{enqueue}(28) \rightarrow [24, 28]$
- $\text{enqueue}(32) \rightarrow [24, 28, 32]$
- $q = \text{dequeue}() \rightarrow \boxed{q = 24}$

$$S + q = 62 + 24 = 86 //$$

2) Postfix Expression

expression: $105 + 606 / * 8 -$

step 1: $105 + = 15$

stack: 15

step 2: $606 / = 10$

stack: 15, 10

step 3: $15 \times 10 = 150$

stack: 150

step 4: $150 - 8 = \boxed{142}$

3. minimum number of enqueue operation

$Q_1 \rightarrow 4$ elements.

$Q_2 \rightarrow 4$ elements.

4). Reverse operation
stack order $O(n)$

5) Post fix expression.

$8 \ 2 \ 3 \wedge 2 \ 3^* + 5 \wedge -$

6) Gate CSE 2007

Infix : $a + b * c \wedge d * e \wedge f$

Postfix:

$a b c d \wedge e f \wedge +$